Mittal School of Business

Project- Predictive Analytics

"Customer Segmentation using K-Means Clustering based on RFM Analysis" (Recency, Frequency and Monetary)

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Submitted by:

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Customer Segmentation using K-Means Clustering on RFM Variables

1. Introduction

Customer segmentation is an important analytical technique used to understand different customer behaviour and design targeted strategies. In this project, customers are segmented based on their purchasing behaviour using **RFM (Recency, Frequency, Monetary**) analysis. The implementation is performed using IBM SPSS Modeler.

2. Objective

The main objectives of this project are:

- To segment customers based on their purchasing behaviour
- To identify high-value, medium-value, and low-value customers
- To support business decisions related to marketing and customer retention

3. Dataset Description

The dataset contains customer-related information, including:

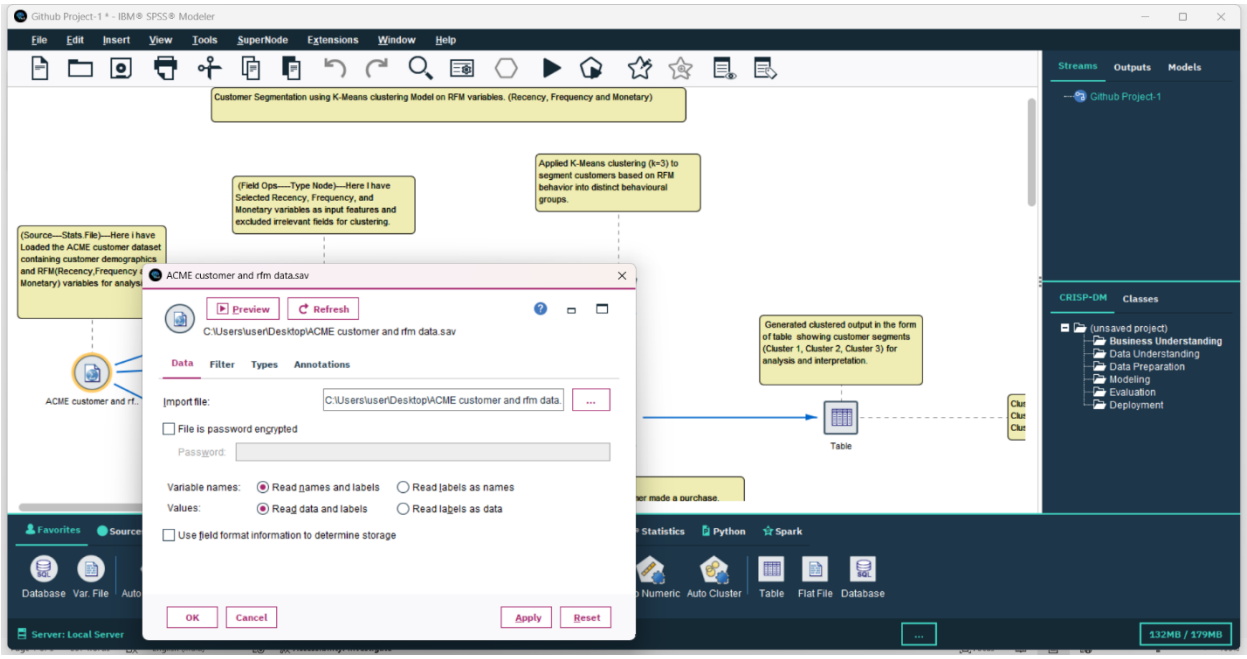
- Customer details and transaction data
- RFM variables:
    - **Recency** – how recently a customer made a purchase
    - **Frequency** – how often the customer purchases
    - **Monetary** – how much money the customer spends

4. Methodology

4.1 Data Import (Source Node)

The dataset was loaded into IBM SPSS Modeler using the Source node.

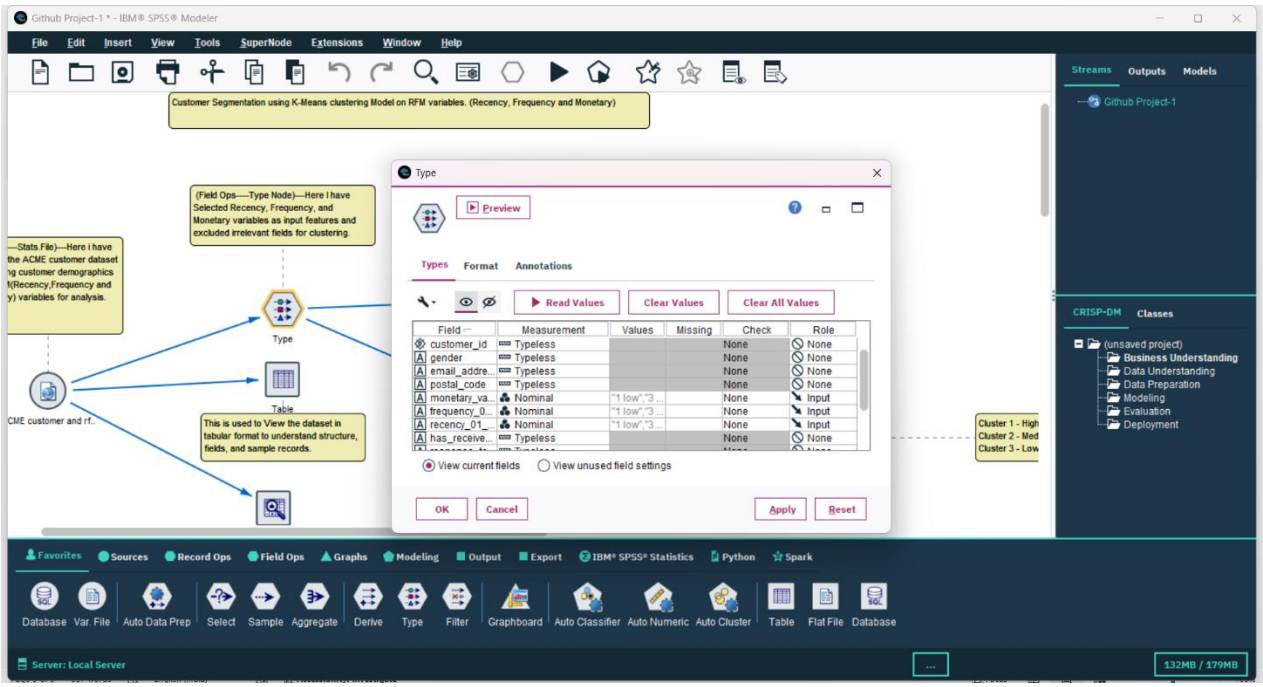
Purpose:
To bring the dataset into the analysis environment for further processing.



4.2 Data Exploration (Table Node)

A Table node was used to view the dataset in tabular format.

Purpose:
To understand the structure of the data, fields available, and sample records.

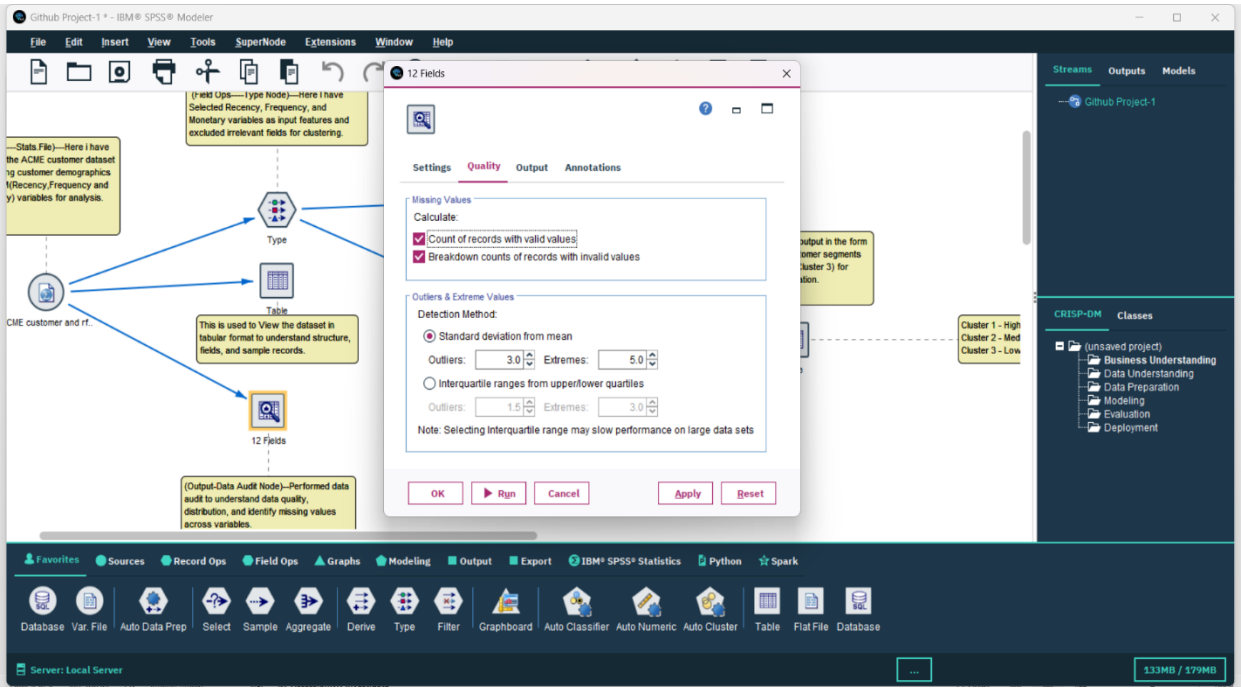


4.3 Data Audit (Data Audit Node)

The Data Audit node was used to analyse data quality.

Purpose:

- To identify missing (null) values
- To understand data distribution
- To check overall data consistency



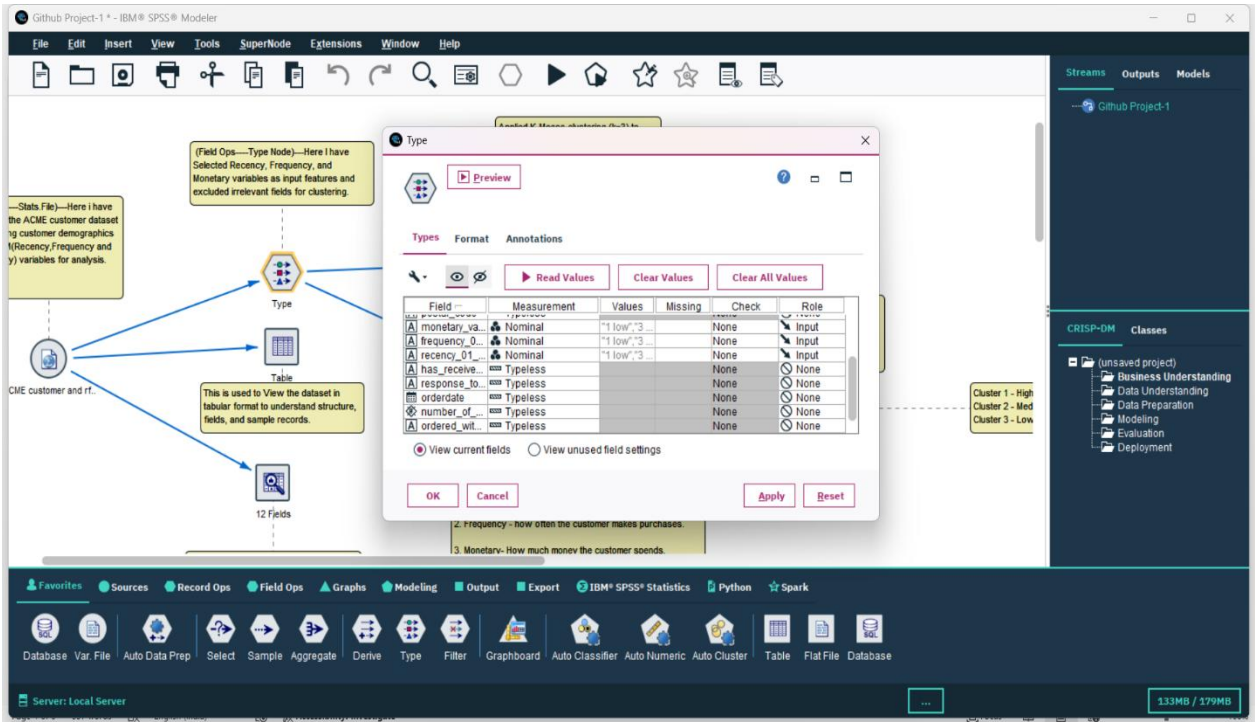
4.4 Feature Selection (Type Node)

The Type node was used to select relevant variables for clustering.

Actions Performed:

- Selected Recency, Frequency, and Monetary as input variables
- Ignored unnecessary fields such as IDs and unrelated attributes

Purpose:
To ensure that only meaningful variables are used in clustering.



4.5 RFM Analysis

The clustering is based on three important variables:

- **Recency:** Indicates how recently a customer made a purchase
- **Frequency:** Indicates how often the customer makes purchases
- **Monetary:** Indicates how much money the customer spends

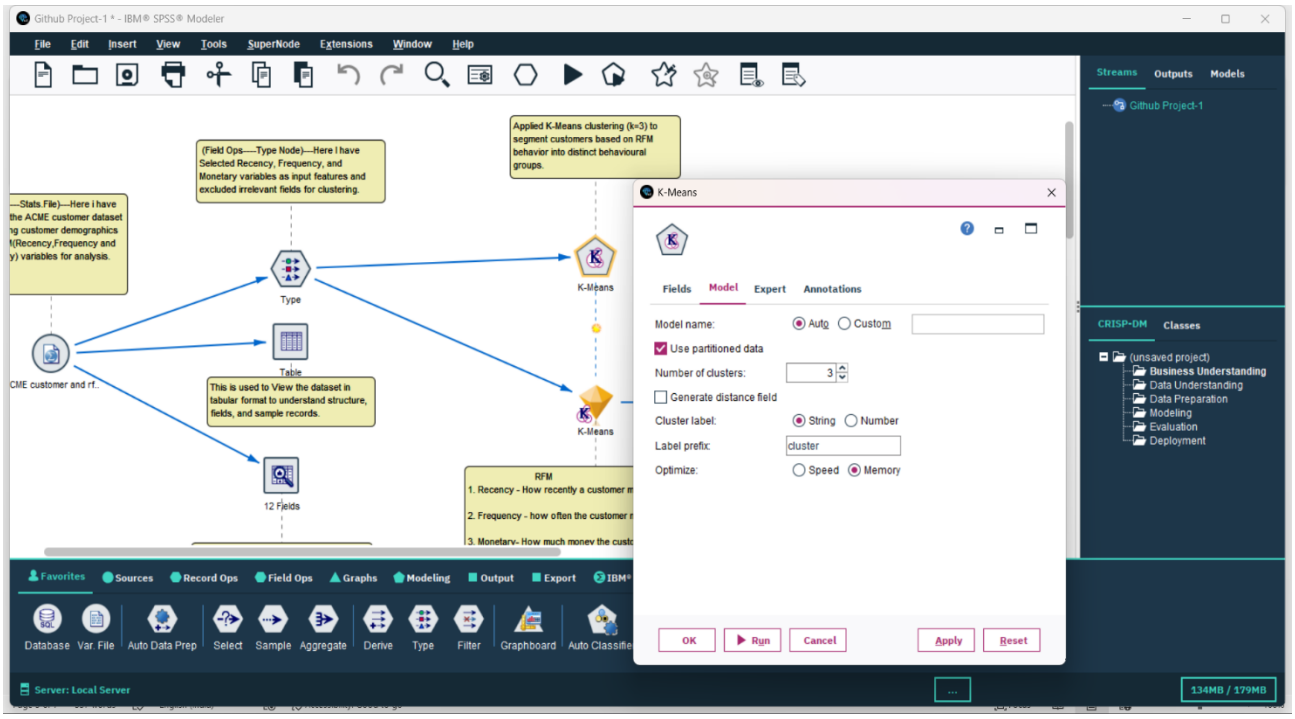
Purpose:
These variables help in understanding customer value and behaviour.

4.6 K-Means Clustering (Model Node)

The K-Means algorithm was applied with k = 3 clusters.

Explanation:
K-Means is an unsupervised learning algorithm that groups similar data points into clusters based on distance.

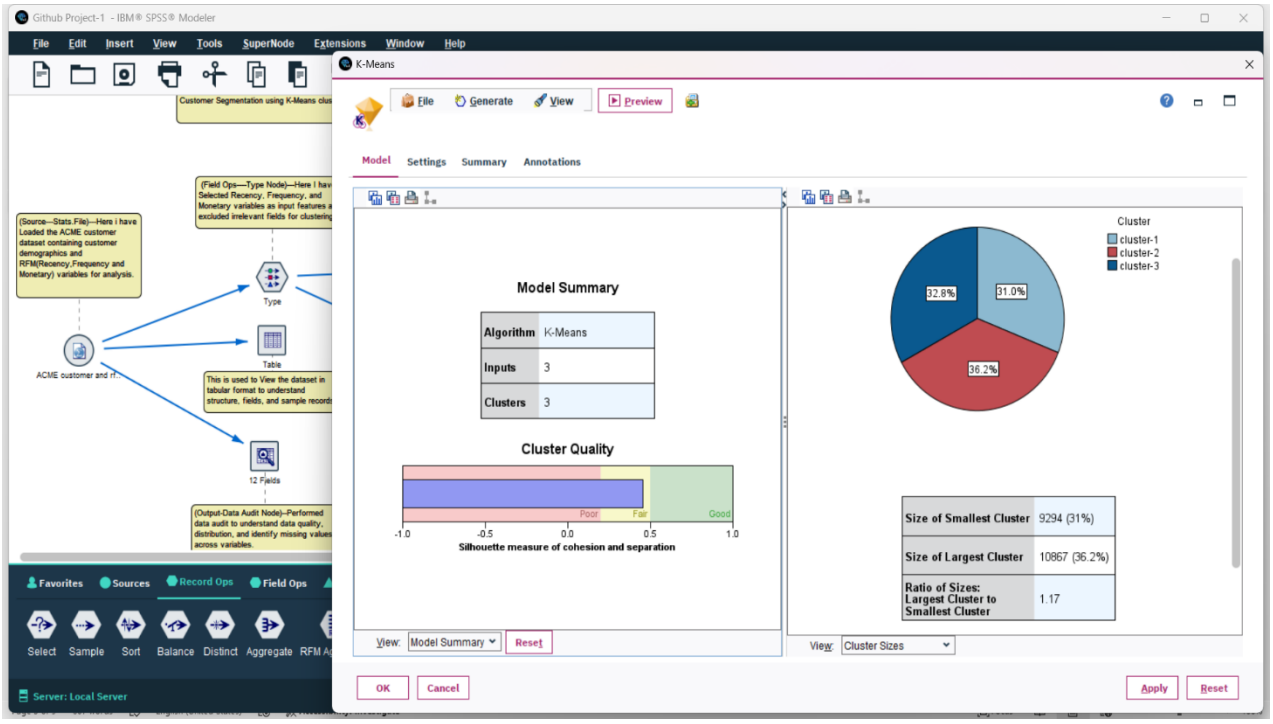
Purpose:
To divide customers into distinct segments based on RFM values.



4.7 Model Execution

The model was executed, and customers were assigned to different clusters.

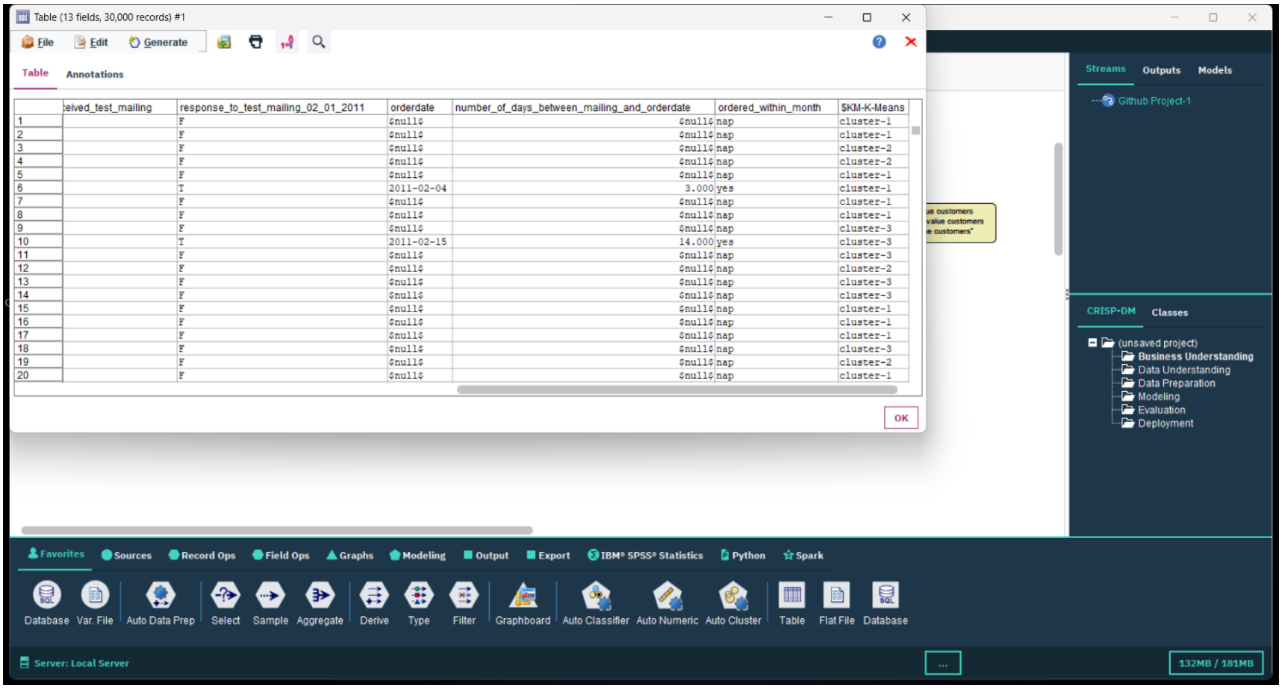
Purpose:
To generate customer segments for analysis.



4.8 Output (Table Node)

A Table node was used to display the final clustered output.

Purpose:
To view cluster assignments for each customer.



	eived_test_mailing	response_to_test_mailing_02_01_2011	orderdate	number_of_days_between_mailing_and_orderdate	ordered_within_month	$KM-K-Means
1		F	$null$	$null$	nap	cluster-1
2		F	$null$	$null$	nap	cluster-1
3		F	$null$	$null$	nap	cluster-2
4		F	$null$	$null$	nap	cluster-2
5		F	$null$	$null$	nap	cluster-1
6		T	2011-02-04	3.000	yes	cluster-1
7		F	$null$	$null$	nap	cluster-1
8		F	$null$	$null$	nap	cluster-1
9		F	$null$	$null$	nap	cluster-3
10		T	2011-02-15	14.000	yes	cluster-3
11		F	$null$	$null$	nap	cluster-3
12		F	$null$	$null$	nap	cluster-2
13		F	$null$	$null$	nap	cluster-3
14		F	$null$	$null$	nap	cluster-3
15		F	$null$	$null$	nap	cluster-1
16		F	$null$	$null$	nap	cluster-1
17		F	$null$	$null$	nap	cluster-1
18		F	$null$	$null$	nap	cluster-3
19		F	$null$	$null$	nap	cluster-2
20		F	$null$	$null$	nap	cluster-1

5. Results

The model created three customer segments:

- **Cluster 1: High-value customers**
  Customer who purchase frequently, recently, and spend more
- **Cluster 2: Medium-value customers**
  Customer with moderate purchasing behaviour
- **Cluster 3: Low-value customers**
  Customer with low frequency and spending

6. Key Observations

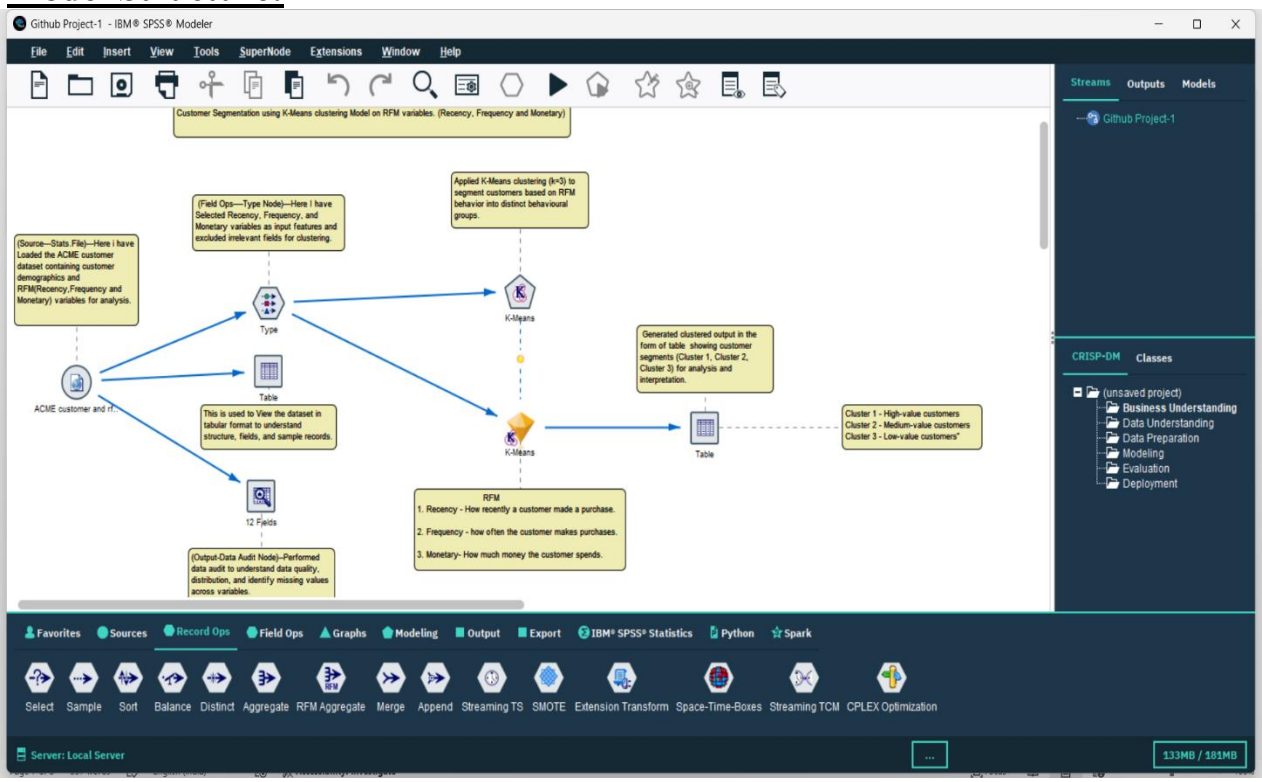
- Customers were successfully grouped based on behaviour
- Some null values were present in the dataset
- RFM variables effectively helped in segmentation

6.1 Model Performance

Model Performance

The clustering model produced three balanced clusters with good distribution across customers. The cluster quality indicates meaningful segmentation based on RFM variables.

Model Structure:



7. Business Implications

- High-value customers can be targeted with loyalty programs
- Medium-value customers can be encouraged to increase purchases
- Low-value customers can be targeted with promotional offers

8. Conclusion

This project demonstrates how K-Means clustering can be used for customer segmentation using RFM analysis. The results provide useful insights for business strategy and decision-making.